COMP 3220 Fall 2023 Assignment 4

Due: Nov. 17, 2023, 11:59 PM

Reading Ch. 18-20, 25 will be helpful for this assignment. Ch. 18-20, 25 were covered in class.

This assignment is implementation-centered. For each question below, you could be asked to provide an explanation for your answer in writing, in a video, or both. You can use either one video to cover all questions requiring video or use multiple short videos each for a question that requires a video submission. If you decide to use one video for all questions, please in your submission indicate the time stamp in your video that answers the question. For example, you can say at time 2 minutes 45 seconds in the video, it is the answer for question 3, and at time stamp 3 minutes 25 seconds in the video, it is the answer for question 4, etc. Please do not upload your video file to Brightspace, but only share with us a link to view your video (youtube link , MS Stream, Google driver, owncloud link, etc, please remember to give proper permission to your shared link to us).

In your source code, please put in comments to help understand your solution.

The textbook provides an "almost complete" implementation for the NextGenPOS case study involving only the happy path of the process sale use case.

1. In the class Payment, there is a private instance variable, shown below

```
private Money amount;
```

We decided to have this attribute of "amount "when we built the domain model. Now, we are at the implementation stage after OOD. Since the implementation language is Java, we quickly realize that we need to Investigate how in Java money (currency) is represented. Money could be a Java class ? or we have to use an alternative if Java does not provide a Money class. Investigate this situation and make a decision on how your implementation handles monetary values.

Note your decision on the Money class may affect other classes in this assignment where the Money class is used.

Write down the result of your investigation and your decision on the Money class. (written submission). Also record a video and explain your handling of this question in the video as well (video submission).

2. In the ProductCatalog class, it contains only 2 ItemID and their associated ProductDescription instances, add at least 3 more to make the program a bit more realistic and interesting. Record a short video showing the changes you make. No written submission is required for this question.

3. All the Java classes provided for this question are not about GUI at all and they are concerned only with the business logic of this case study and the current use case. Moreover, there is no Java "main" method to bring all the classes alive and to test that they all work collaboratively to realize the process sale use case. Since most of the students in the class indicated that not sufficient Java GUI was covered in COMP 2120, in this question, you need to simulate the interaction between the cashier and the system, using a text-based console window (as some of you did in your Assignment 1). Write the necessary class(es) of your choice to simulate this text-based "GUI" instead of putting all the "GUI" code in the Java main method. However, if you are apt at Java GUI, you could go ahead to use real Java GUI.

   You need to think about how your "GUI" class interacts with all the software classes in your business logic. In other words, make a decision on how you would apply the controller pattern.

   In your solution for this question, please write down the design of your GUI (to help us understand your code) and how you use the controller pattern for this question. Also record a video to verbally explain your design of "GUI", your application of the controller pattern, and how your Java main method weaves everything together to demonstrate the happy path of the process sale use case. Both written and video submissions are needed for this question.

   All the code you have had so far after finishing questions 1 to 3 is referred to as the baseline code V1 in this assignment, and they should demonstrate a working Java program realizing the happy path of the process sale use case. Zip all your files and name the zipped file baseline-code-v1-firstname-lastname.zip and submit the zip file.

4. The baseline code V1 so far does not consider tax rate at all when calculating the total for a sale. Your NextGenPOS system will be used in Ontario for now and is expected to be sold to Quebec businesses as well in the near future. The Ontario Government provides an API to let all businesses know the tax rate for each item they sell. An extremely simplified class simulating the Ontario tax rate API is provided. Using the appropriate pattern(s), make changes/add additional code to the relevant part of your existing baseline code V1 so that every time you calculate a subtotal/total, the appropriate tax rate for Ontario is applied. It is expected the Java main method will probably have to be modified as well. After finishing this question, you obtain a working program that uses the Ontario tax rate API.

   Explain in writing first what pattern should be used in this case to handle the Ontario tax rate API, then explain what class/method is affected in your baseline code. Also, record a video to explain the changes you made to the baseline code and how the whole program works.

   All the code you need to make question 4 work is referred to as the baseline code V2 in this assignment, and they should demonstrate a working Java program realizing the happy path of the process sale use case using Ontario tax rate API. Zip all your files and name the zipped file baseline-code-v2-firstname-lastname.zip and submit the zip file as well.

(A hint to the main method: how does the main method know it should use the Ontario Tax rate API or the Quebec Tax rate API, the textbook discussed a similar situation.)

5. This question is a continuation of the previous question. There is now a buyer of your NextGenPOS system from Quebec. The Quebec Government provides its own API to let all businesses know the tax rate for each item they sell in Quebec. An extremely simplified class simulating the Quebec tax rate API is provided. What changes would happen to your code in your baseline code V2. Make changes to your baseline code V2 code so that every time you calculate a subtotal/total, the appropriate tax rate for Quebec is applied. Explain what pattern should be used in your solution and what changes to your baseline code V2. Explain those changes you made in a video. Also, If you make any changes to the main method you previously have in question 4 for this question, please explain that change in the video as well.

   All the code you need to make question 5 work is referred to as the baseline code V3 in this assignment, and they should demonstrate a working Java program realizing the happy path of the process sale use case using the Quebec tax rate API. Zip all your files and name the zipped file baseline-code-v3-firstname-lastname.zip and submit the zip file as well.

6. For the 4 different kinds of visibility discussed in the textbook (Ch. 19), find one example for each kind. Record a short video to show each one of them.

   Keep all your Java files and they may be needed for the next assignment.

Question	Written Submission	Video Submission (link)
1	Yes	Yes
2	no	Yes
3	Yes	Yes
	Submit baseline code V1 zip file	
4	Yes	Yes
	Submit baseline code V2 zip file	
5	Yes	Yes
	Submit baseline code V3 zip file	
6	No	Yes